Auditable Hierarchical Inference with Multimodal Evidence and Bayesian Active Inquiry

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Abstract

Most multimodal classifiers assume that all useful evidence has already been collected. Clinical inquiry is sequential: passive observations should determine which question is useful next, while missing evidence and unsafe outputs require explicit containment. We present a synthetic, four-examination-inspired evaluation framework that separates objective multimodal inference from active inquiry. An objective-inference model maps synthetic tongue, bilateral multisite multipressure pulse, and acoustic observations to a calibrated distribution over eight syndrome families. A provenance-bearing Knowledge Core and a training-only answer model then support Bayesian narrowing over 33 fine syndrome terms using 25 typed questions; seven safety questions are isolated from the diagnostic posterior. Fine terms influence objective phenotypes only through weak shared symptom factors, and an objective-only leakage probe constrains direct term recovery.

In a single, preregistered 600-case synthetic confirmation, the objective-inference model achieved 87.0% family accuracy. Cost-sensitive expected information gain (cost-EIG) increased family accuracy to 95.5% and exact fine-term accuracy to 42.5%, compared with 92.7% and 33.0% for the tested unoptimized fixed-order implementation. Selective family accuracy was 96.9% at 95.3% coverage. Cost-EIG did not reduce question count: it asked 9.795 questions on average, versus 9.352 for fixed order. Missing-audio routing retained 94.0% family accuracy. A workflow invariant mandates pre-inquiry abstention when all objective modalities are absent. Permuting the fine-term answer-likelihood mapping reduced family accuracy to 78.7%, the largest degradation among the tested synthetic stress conditions. Five of six frozen checks passed; the question-efficiency hypothesis failed. All numerical comparisons are descriptive because the one-time artifact retained aggregate, not paired case-level, outputs. These results establish a reproducible synthetic workflow result, not clinical diagnostic or treatment validity.

1 Introduction

The practical question addressed here is not whether a classifier can make a terminal prediction after receiving a complete record. It is whether a system can (i) form an auditable prior from objective evidence, (ii) choose which structured observation to acquire next, (iii) preserve uncertainty across a coarse-to-fine label hierarchy, and (iv) stop or abstain when evidence is insufficient. This distinction matters when questions impose burden, answers can be missing or contradictory, and the knowledge used to guide acquisition can itself be wrong.

This study operationalizes the problem in a bounded synthetic setting. The objective-inference module first predicts one of eight syndrome families from objective modalities. It then narrows a posterior over 33 prespecified fine syndrome terms using typed questions and a Bayesian answer model estimated only from training cases. The Knowledge Core defines terminology, family

membership, symptom relations, question semantics, and provenance, but is treated as a fallible component rather than an oracle. Language generation is outside the present experiment: the confirmation uses structured answers so that model, policy, answer-model, and knowledge failures remain identifiable.

The evaluated contributions are fourfold:

1. a hierarchical generator in which fine terms affect objective phenotypes through shared symptom factors, together with a family-conditional leakage probe that limits direct fine-term recognition;
2. an objective multimodal inference model with explicit routing for missing modalities and mandatory abstention when all objective evidence is absent;
3. an auditable Bayesian acquisition layer that compares cost-EIG with fixed, random, candidate-contrast, and unpenalized EIG policies under the same question contract and stopping budget; and
4. a one-time frozen confirmation with separate hypothesis verdicts and targeted stress tests for answer, modality, and knowledge failures.

The paper does not claim clinical validity, treatment recommendation, expert equivalence, language-model superiority, or superiority of the full pulse tensor over reduced projections. Those require separate studies.

2 Related Work and Positioning

Active feature acquisition studies how a predictor can purchase or query previously unobserved features at inference time while trading predictive value against acquisition burden. Model-based approaches have used generative surrogates to estimate the value of unobserved features [1], while more recent work has studied nongreedy acquisition with an acquisition-conditioned oracle [2]. The proposed workflow does not introduce EIG or active feature acquisition. Its contribution is instead an auditable hierarchical integration in which a multimodal family prior, a train-estimated answer model, a provenance-bearing knowledge resource, a safety firewall, and stopping or abstention rules have separately testable authority.

Adaptive inquiry has also been studied in TCM-oriented syndrome differentiation. Li et al. constructed a consultation knowledge base and evaluated multi-round syndrome inquiry using diagnostic accuracy, question quality, and question count [3]. Our study differs by separating objective multimodal inference from typed Bayesian updating, withholding simulator-private answer probabilities from the policy, and injecting failures at declared module boundaries. It is complementary rather than a claim of first adaptive TCM questioning.

The pressure–space–time pulse tensor is motivated by sensing work that moves beyond single-point waveforms. Wearable pressure arrays have measured Cun, Guan, and Chi locations and reconstructed temporal–spatial pulse maps [4]. Our renderer adopts the general engineering premise that site, transverse extent, applied pressure, and time should remain explicit, but it does not reproduce that hardware or inherit clinical validity from it. Likewise, modality-wise dropout has prior use as a strategy for robustness to missing multimodal inputs [5]. Our contribution is the combination of dropout training with an explicit availability router and a pre-inquiry abstention contract, not the invention of modality dropout.

Because the system may abstain and exposes posterior confidence, calibration and selective prediction are part of the evaluation rather than incidental metrics. Modern neural classifiers

can be miscalibrated [6], and selective prediction formalizes the accuracy–coverage trade-off under a reject option [7]. We therefore report NLL, Brier score, ECE, coverage, and selective accuracy alongside top-k accuracy. The present study does not claim a new calibration or selective-prediction algorithm.

3 Problem Definition and System Boundary

Let $T \in \{1, \ldots, 33\}$ denote a fine syndrome term and $S = g(T) \in \{1, \ldots, 8\}$ its syndrome family. Let Z be 32 shared atomic symptom factors, U a nine-factor patient phenotype, and X the objective multimodal observation. Synthetic cases follow

$$T \to S, \qquad T \to Z \to U \to X. \tag{1}$$

This construction allows weak family-internal heterogeneity without loading a fine-term identifier directly into an image, pulse trace, or audio waveform. Fine-term labels are balanced within each split by a shuffled round-robin rule. A family-conditional objective-only probe rejects a generator if fine-term accuracy exceeds the corresponding within-family chance level by more than 0.10.

The objective-inference module receives only X at inference. During training it is supervised by the family label S and low-weight shared-factor targets Z; it receives neither T, Knowledge Core text, question answers, nor a disease label. The inquiry module receives the objective family distribution and typed question answers. It does not receive raw dialogue. The evaluated system therefore ends at synthetic syndrome-family and fine-term resolution; the disease field is null, and no treatment action is defined.

The architecture separates five authorities: objective representation, probabilistic inference, question selection, safety handling, and human-facing language. Only the first three are quantitatively evaluated here. A future language model may realize an already selected question and map a response to a typed distribution, but it may not create labels, write posterior probabilities, change stopping rules, or recommend treatment.

4 Objective Multimodal Evidence

4.1 Tongue, pulse, and acoustic channels

The evaluated full model uses three objective channels: a synthetic tongue image, a standardized pulse tensor, and task-controlled acoustic observations. The pulse record is

$$X^{\text{pulse}} \in \mathbb{R}^{2\times3\times3\times7\times512}, \tag{2}$$

corresponding to two hands, Cun–Guan–Chi sites, superficial–middle–deep pressure levels, seven transverse sensors, and eight seconds sampled at 64 Hz. Each record also contains acquisition pressure, quality, an artifact mask, and 16 continuous properties describing depth response, rate, irregularity, force, transverse support, spatial extent, smoothness, tension, pressure persistence, stability, hollowness, edge firmness, upstroke, decay, and pause behavior.

Twenty-eight pulse qualities are overlapping semantic projections of these continuous properties rather than mutually exclusive fixed templates. The renderer explicitly distinguishes, for example, floating from deep by cross-pressure response, fine from large by transverse support, and deficient from replete by absolute force and persistence. Dashed visualization denotes uncertain or intermittent detection rather than serving as a class code. The runtime renderer consumes anonymous state identifiers and a hashed ontology.

An auxiliary 28-quality pulse classifier with continuous-property regression was implemented as a representation audit. Importantly, the final objective-inference model used random pulse-encoder initialization. The confirmation therefore does not support a transfer-learning or full-versus-reduced pulse-representation claim. The supported pulse result is narrower: an early family-template shortcut was detected, the generator was remediated with overlap controls and registered perturbations, and the final multimodal model was evaluated after that audit.

The acoustic interface produces canonical mono PCM WAV data and a hashed manifest for fixed phrase, sustained vowel, cough, and quiet-breathing tasks. Controls span phonation, intensity, timing, resonance, respiration, cough force and wetness, wheeze, and crackle; gain, bandwidth, distance, room noise, and reverberation are explicit nuisances. Audio was included in the final full model. It is an engineering channel for the synthetic experiment and is not evidence of clinical auscultation validity.

4.2 Objective-inference training and availability routing

The passive model encodes the available objective modalities and emits eight family logits. Model selection uses development negative log-likelihood (NLL). The selected full model is trained with registered pulse perturbations and modality dropout. Calibration, stopping, abstention, conflict thresholds, and the inference router are frozen before confirmation access.

At inference, an availability router chooses among frozen full, tongue–pulse, pulse-only, audio-only, and tongue-only experts. It does not average logits from experts trained for incompatible availability states. If every objective modality is absent, the workflow abstains before inquiry; questions are not allowed to manufacture an objective prior.

5 Knowledge Core and Active Inquiry

5.1 Knowledge and question contracts

The Knowledge Core is fixed across all evaluated conditions and is identified by source-registry, term-record, symptom-link, pulse-record, candidate-subset, and question-contract hashes. The frozen experiment uses a bounded synthetic subset of 33 fine terms rather than treating all 2,060 extracted terms as flat classes. This subset is internally fixed for reproducibility but has not been validated by an external TCM expert.

Each question declares an identifier, one atomic symptom axis, answer schema, cost, sensitivity, provenance, and status. It contains neither fine-term applicability flags nor private generator probabilities. Of 32 axes, 25 are eligible for Bayesian narrowing. Seven potential warning-sign questions form an independent safety screen. Safety answers can trigger a recorded action but are prohibited from changing the fine-term posterior or locking a syndrome result.

5.2 Bayesian update and acquisition

Let $H_t = \{(q_i, a_i)\}_{i=1}^{t}$ be the structured inquiry history. The passive family distribution and predicted shared factors induce the initial fine-term belief

$$p_0(T \mid X) = \frac{p_0(g(T) \mid X)\, p_\phi(T \mid g(T), \widehat{Z}(X))}{\sum_{T'} p_0(g(T') \mid X)\, p_\phi(T' \mid g(T'), \widehat{Z}(X))}. \tag{3}$$

For a typed answer a to question q, the posterior update is

$$p_{t+1}(T) \propto p_t(T)\, \widehat{p}_{\text{train}}(a \mid T, q), \tag{4}$$

where the answer likelihood uses Laplace smoothing and training cases only. A missing answer leaves the posterior unchanged. The simulator's private answer probabilities are never exposed to the acquisition policy.

The expected information gain is

$$\mathrm{EIG}(q \mid H_t) = H[p_t(T)] - \mathbb{E}_{a \sim \widehat{p}_{\mathrm{train}}(a \mid q, H_t)} H[p(T \mid H_t, q, a)]. \tag{5}$$

The proposed policy selects

$$q_t^* = \arg\max_{q \notin H_t} \left\{ \mathrm{EIG}(q \mid H_t) - \lambda c(q) - \gamma r(q, H_t) \right\}, \tag{6}$$

where $c(q)$ is burden and $r(q, H_t)$ penalizes repetition or contextual inappropriateness. The budget is 12 questions. Development selection fixed a confidence threshold of 0.65 and an abstention threshold of 0.60.

6 Experimental Protocol

The study uses 2,400 synthetic training cases, 600 development cases, and a 600-case confirmation split. The confirmation split was accessed once after the generator, models, router, question contract, answer estimator, policies, thresholds, endpoints, and claim gates were frozen. Rerunning confirmation or post-confirmation tuning is prohibited.

The primary objective-inference endpoint is eight-family top-1 accuracy. The inquiry module reports family and exact fine-term top-1 accuracy, top-3 recall, NLL, Brier score, expected calibration error (ECE), mean questions, coverage, and selective accuracy. Exact fine-term accuracy is a synthetic resolution endpoint, not a clinical diagnosis endpoint. The prespecified policy comparison holds the prior, answer model, question set, stopping rules, and budget constant and varies only question selection:

- fixed order and random order;
- candidate contrast, which targets disagreement among leading candidates;
- greedy EIG; and
- cost-EIG, the proposed policy.

This control has an important limitation. Every policy uses the same confidence and low-information stopping rules. For the tested fixed policy, the next question is the next index in a predetermined, unoptimized order. If that specific question falls below the information threshold, the episode stops; the policy does not scan later fixed questions for a more informative one. Consequently, the fixed condition is neither a conventional fixed-length questionnaire nor a development-optimized static order. The valid comparison is with this tested fixed-order implementation, not with all well-designed static questionnaires. An equal-budget curve and a train-optimized static ordering were not evaluated.

Question count is reported as the prespecified burden proxy. Although cost-EIG uses declared per-question costs inside its selection score, the frozen report does not retain the realized cumulative cost $C(H_T) = \sum_{t=1}^{T} c(q_t)$. "Cost-sensitive" therefore names the selection objective; it is not evidence that the resulting policy minimized realized burden. The relevant observed comparison is that cost-EIG retained EIG's fine-term accuracy while using fewer questions than EIG, but more than the tested fixed order.

Stress tests cover missing answers, contradictory answers, missing-audio routing, absence of all objective modalities, absent answer-likelihood information, fine-term likelihood-row permutation, and answer-model misspecification. Six frozen checks test objective-inference family accuracy, cost-EIG family accuracy, fine-term advantage over fixed order, question-count efficiency, missing-audio accuracy, and all-objective-missing abstention. Each hypothesis retains its own verdict; improved accuracy cannot be used to relabel a failed efficiency test.

6.1 Statistical interpretation

The confirmation is a single deterministic evaluation on 600 synthetic cases. The frozen artifact stores aggregate metrics but not paired case-level predictions or complete acquisition trajectories. Paired bootstrap intervals, McNemar tests, paired permutation tests, and uncertainty intervals for policy differences cannot be reconstructed from that artifact. We do not reopen the consumed confirmation split to add a favorable post hoc analysis. Results are therefore reported as descriptive estimates under one generator seed and one frozen environment. In particular, the 0.7-point rounded family-accuracy difference between cost-EIG and EIG corresponds to only four cases and is not interpreted as reliable policy superiority. A separately registered replication is required for inferential comparisons.

7 Results

7.1 Development audits

The original pulse generator was rejected after a simple probe achieved 94.2% family accuracy, revealing a family-template shortcut. Following overlap control and registered perturbation augmentation, pulse-only development accuracy fell to 79.7%, consistent with removal of an overly deterministic shortcut rather than with a performance optimization objective. The selected dropout-trained full model achieved 89.2% family accuracy, NLL 0.288, Brier score 0.160, and ECE 0.039 on complete development inputs. Its direct missing-audio performance failed the registered gate, motivating the frozen availability router rather than an unregistered retraining cycle.

On development data, cost-EIG achieved 95.3% family accuracy, 43.8% exact fine-term accuracy, and 9.65 mean questions. The shared-factor compatibility term changed the first question in 8.5% of cases, supporting a modest personalization effect rather than a claim that passive evidence fully determines the inquiry trajectory.

7.2 Frozen confirmation

The objective-inference model achieved 87.0% family accuracy, 99.7% family top-3 recall, NLL 0.320, Brier score 0.181, and ECE 0.022. Table 1 reports all frozen policy comparisons. Cost-EIG and unpenalized EIG tied at 42.5% exact fine-term accuracy. Cost-EIG asked fewer questions than EIG, candidate contrast, and random order, but more than fixed order. Its main advantage over fixed order was resolution quality: +2.8 percentage points for family accuracy and +9.5 points for exact fine-term accuracy.

Cost-EIG reached 95.5% family accuracy and 96.9% selective family accuracy at 95.3% coverage. However, its 9.795 mean questions exceeded fixed order by 0.443. The prespecified question-efficiency claim therefore failed. Furthermore, 56.0% of high-family-confidence cost-EIG cases still had the wrong fine term. Family confidence must not be interpreted as fine-term confidence.

Policy	Family acc. (%)	Fine acc. (%)	Fine top-3 (%)	Fine NLL	Questions mean	Sel. family acc. (%)
Fixed order	92.7	33.0	78.2	1.562	**9.352**	95.6
Random order	92.0	31.5	77.2	1.598	10.058	93.9
Candidate contrast	93.7	42.0	84.2	1.444	10.643	94.7
Greedy EIG	94.8	**42.5**	**86.8**	1.398	10.457	**96.9**
Cost-EIG	**95.5**	**42.5**	85.8	**1.388**	9.795	**96.9**

Table 1: One-time synthetic confirmation ($n = 600$). Selective family accuracy is computed among non-abstained cases; coverage was 94.8%, 95.0%, 97.7%, 95.8%, and 95.3% in table order. Values are descriptive point estimates. Bold marks the best value, allowing rounded ties; it does not denote statistical significance.

Cost-EIG and EIG had identical fine-term top-1 accuracy. Relative to EIG, cost-EIG reduced mean question count from 10.457 to 9.795 while changing family accuracy by four cases. Because cumulative declared cost was not retained, this supports count reduction relative to EIG, not a demonstrated improvement in total accuracy–burden trade-off.

7.3 What the experiment does not isolate

The development personalization audit changed the first selected question in 8.5% of cases, but the confirmation did not compare the full prior with a family-only prior, removal of shared-axis personalization, a uniform prior, or an oracle family prior. The observed improvement can therefore reflect a combination of the coarse family prior, Bayesian adaptation to previous answers, and the question policy. It does not isolate how much the rich multimodal representation caused the question sequence to improve. This is the central mechanism question for a future replication, not a result of the present confirmation.

7.4 Failure injection and hypothesis verdicts

Missing answers and the frozen missing-audio route caused moderate degradation, whereas contradictory answers caused a larger calibration and resolution loss (Table 2). When all objective modalities were absent, the system was specified to abstain before inquiry. This is a tested workflow invariant, not a case-derived empirical rate. Removing the answer-likelihood information reduced the system to its passive prior and stopped inquiry for low information. The "knowledge corrupted" stress specifically permuted the fine-term rows of the answer-likelihood matrix; it did not corrupt every component of the Knowledge Core. This permutation reduced family accuracy to 78.7%, exact fine-term accuracy to 20.3%, and selective family accuracy to 82.5%, while family ECE rose to 8.6%. Fine-term likelihood-map corruption therefore produced the largest degradation among the tested synthetic stresses.

Five of the six frozen checks passed: objective-inference family accuracy, cost-EIG family accuracy, fine-term advantage over the tested fixed order, missing-audio family accuracy, and the all-objective-missing abstention invariant. The first four are case-derived empirical checks; the fifth is an implementation-contract check. Question-count efficiency failed. The safety-to-posterior firewall was exercised in development tests, not scored as a confirmation-set clinical outcome. No model, threshold, claim gate, or confirmation result was changed after access.

Condition	Family acc. (%)	Fine acc. (%)	Questions mean	Coverage (%)	Sel. family acc. (%)
Nominal cost-EIG	95.5	42.5	9.795	95.3	96.9
Missing answers	93.8	38.2	10.322	96.2	96.0
Contradictory answers	91.2	32.8	9.645	94.3	93.6
Missing audio route	94.0	41.2	10.268	92.7	96.0
Likelihood absent	87.0	22.3	0.000	87.7	92.2
Term-map permutation	78.7	20.3	6.455	89.7	82.5
Answer model misspec.	93.0	41.5	4.410	94.2	95.6

Table 2: Targeted, descriptive confirmation stress tests. Each row changes one declared interface relative to nominal cost-EIG. "Term-map permutation" permutes fine-term answer-likelihood rows and is not arbitrary Knowledge Core corruption. All-objective-missing is omitted because it was evaluated as a workflow invariant rather than a case-derived metric.

Check	Evidence source	Verdict	Interpretation
Objective-inference family accuracy	600 confirmation cases	Pass	Empirical
Cost-EIG family accuracy	600 confirmation cases	Pass	Empirical
Fine-term gain vs. tested fixed	Paired split, aggregate retained	Pass	Descriptive
Question-count efficiency	600 confirmation cases	Fail	Empirical
Missing-audio family accuracy	Routed confirmation cases	Pass	Empirical
All objective inputs missing	Evaluator workflow guard	Pass	Invariant
Safety-to-posterior firewall	Development integration tests	N/A	Invariant

Table 3: Evidence type is reported separately from the pass/fail status. A software invariant is not presented as an additional case-derived clinical outcome. N/A denotes a development-tested contract that was not one of the six frozen checks.

8 Discussion

The strongest result is not that cost-EIG dominates every baseline on every metric. It does not. Rather, the frozen system demonstrates that a passive multimodal prior and a structured knowledge-guided inquiry layer can be evaluated as separate modules, with measurable coarse-to-fine gains and attributable failure modes. Cost-EIG improved exact term resolution over fixed and random orders and used fewer questions than the other adaptive policies, but the tested fixed policy remained the cheapest by raw count. Because that policy used an unoptimized order and stopped when its next item failed the shared information threshold, these data show superiority to one particular fixed implementation, not to optimized or equal-length questionnaires. The cost term changed the adaptive-policy trade-off without establishing global burden superiority.

The hierarchical endpoint is also essential. Objective inference was already useful at the family level, while the 33-way exact term task remained difficult even after inquiry. Reporting only family accuracy would hide substantial residual fine-term uncertainty; reporting only exact-term accuracy would hide the useful coarse prior. The paired endpoints make the intended division of labor explicit: objective evidence establishes a family distribution, and inquiry attempts within- and cross-family refinement.

Failure injection changes the novelty claim from generic "Bayesian questioning" to auditable acquisition under fallible inference resources and missing modalities. The workflow contract contained total objective absence, and the empirical missing-audio route preserved performance. Permuting the fine-term answer-likelihood map produced confident harm. A deployable successor would re-

quire provenance-aware trust weights, contradiction monitoring, and a knowledge-disagreement abstention rule validated independently of this test set.

The defensible systems novelty is the combination of explicit authority boundaries, hierarchical uncertainty, leakage control, frozen acquisition comparisons, and interface-specific failure injection. EIG, active feature acquisition, calibration, selective prediction, and adaptive TCM inquiry all have prior literature. Accordingly, the contribution is an auditable architecture and evaluation protocol connecting multimodal evidence to active inquiry, not a new general acquisition algorithm.

The 28-quality pulse representation adds an engineering contribution by encoding depth, transverse support, pressure response, and temporal structure in a counterfactually controllable tensor. The present evidence supports the renderer and shortcut-audit workflow, not a clinical claim about the 28 qualities and not a superiority claim over a 1-D waveform. Those questions remain separate ablation and human-validation studies.

9 Limitations

All cases, modalities, answers, and labels are synthetic and share a designed causal structure. This can overstate cross-modal coherence and makes the reported accuracies unsuitable for clinical extrapolation. The 33 terms and eight families are internally frozen research labels, not an expert-adjudicated reference standard. Some fine distinctions may be near-synonymous or underdetermined by the 25-question contract.

The one-time confirmation provides a disciplined held-out estimate but only one synthetic split; no paired confidence interval, resampling analysis, external generator, or prospective cohort is reported. The stopping and abstention thresholds are calibrated primarily for the family endpoint, and the high rate of fine-term error among family-confident cases shows that they should not be reused as fine-term certainty thresholds.

Baseline fairness is incomplete. The fixed order was neither optimized on the training/development data nor evaluated at every equal question budget, and its shared low-information stopping rule can terminate before a later fixed question. The study reports question count rather than accumulated question cost. It also lacks an all-questions upper bound, independent generator seeds, and a parameter- shift environment. These omissions limit both the adaptive-policy and cost-sensitive claims.

The confirmation does not isolate the causal contribution of objective multimodal personalization to acquisition. Family-only, no-shared-axis, uniform- prior, and oracle-family ablations were not frozen for the consumed split. The 8.5% development first-question change demonstrates that personalization can alter behavior, but not that it explains the terminal gain.

The pulse auxiliary model was not transferred into the final model, and the full tensor was not compared with reduced projections under equal budgets. The audio generator has not been validated against real recordings. No natural-person TCM expert reviewed the current ontology, pulse mapping, or question contract. This does not invalidate the synthetic systems result, but it prohibits clinical interpretation.

Finally, no free-text patient, LLM question realization, or uncertain answer extractor was evaluated. The proposed language boundary is an architectural contract only. A future language-layer study should freeze the structured policy, compare typed oracle and model-based extraction, propagate extraction probabilities rather than hard labels, and measure semantic error independently from diagnostic resolution.

9.1 Registered replication priorities

A new confirmation split, rather than reuse of the consumed one, should evaluate (i) a development-optimized static order and fixed-length budgets from zero to 12 questions; (ii) accuracy–question and accuracy–cumulative-cost curves; (iii) full, family-only, no-shared-axis, uniform, and oracle-family priors; (iv) an all-questions upper reference; and (v) independent generator seeds and a parameter-shifted environment. Per-case predictions, complete trajectories, and cumulative cost should be retained under a preregistered analysis plan so that paired bootstrap intervals and paired tests can be computed without post hoc test access.

10 Reproducibility and Claim Control

The final release records hashes for the development protocol, confirmation plan, model artifacts, Knowledge Core inputs, and confirmation outputs. The 600-case confirmation access marker is consumed and rerun is prohibited. The release status is *final synthetic confirmation with partial hypothesis support*; clinical claims are explicitly unauthorized. External expert validation is not a completion requirement for this software release, but remains a prerequisite for any clinical or traditional-medicine validity claim.

The accompanying repository contains the synthetic generator, frozen configurations, question contract, evaluation code, aggregate confirmation report, release hashes, and a scoped validation suite. The declared artifact validation suite passed all 69 tests and serves as the reproducibility gate for this study. No human participants, patient records, or biological specimens were used. Consequently, the current study creates no human-subject clinical evidence and makes no treatment recommendation.

11 Conclusion

This work reframes synthetic multimodal classification as hierarchical evidence acquisition. In the frozen confirmation, objective evidence supplied a useful eight-family prior, structured Bayesian inquiry improved family and fine-term resolution relative to the tested fixed and random implementations, and empirical missing-audio routing remained robust. Pre-inquiry abstention under total objective absence and safety-screen separation are implementation contracts with development-test support, rather than additional confirmation-set clinical outcomes. The same study rejected the claim that cost-EIG asks fewer questions than the tested fixed order; it did not measure cumulative cost or isolate the multimodal personalization mechanism. Fine-term answer-likelihood mapping permutation was the most damaging tested stress. The appropriate conclusion is therefore narrow: the system supports auditable synthetic coarse-to-fine inference and failure decomposition, not clinical diagnosis, treatment guidance, general question-efficiency superiority, optimized-static-baseline superiority, or LLM superiority.

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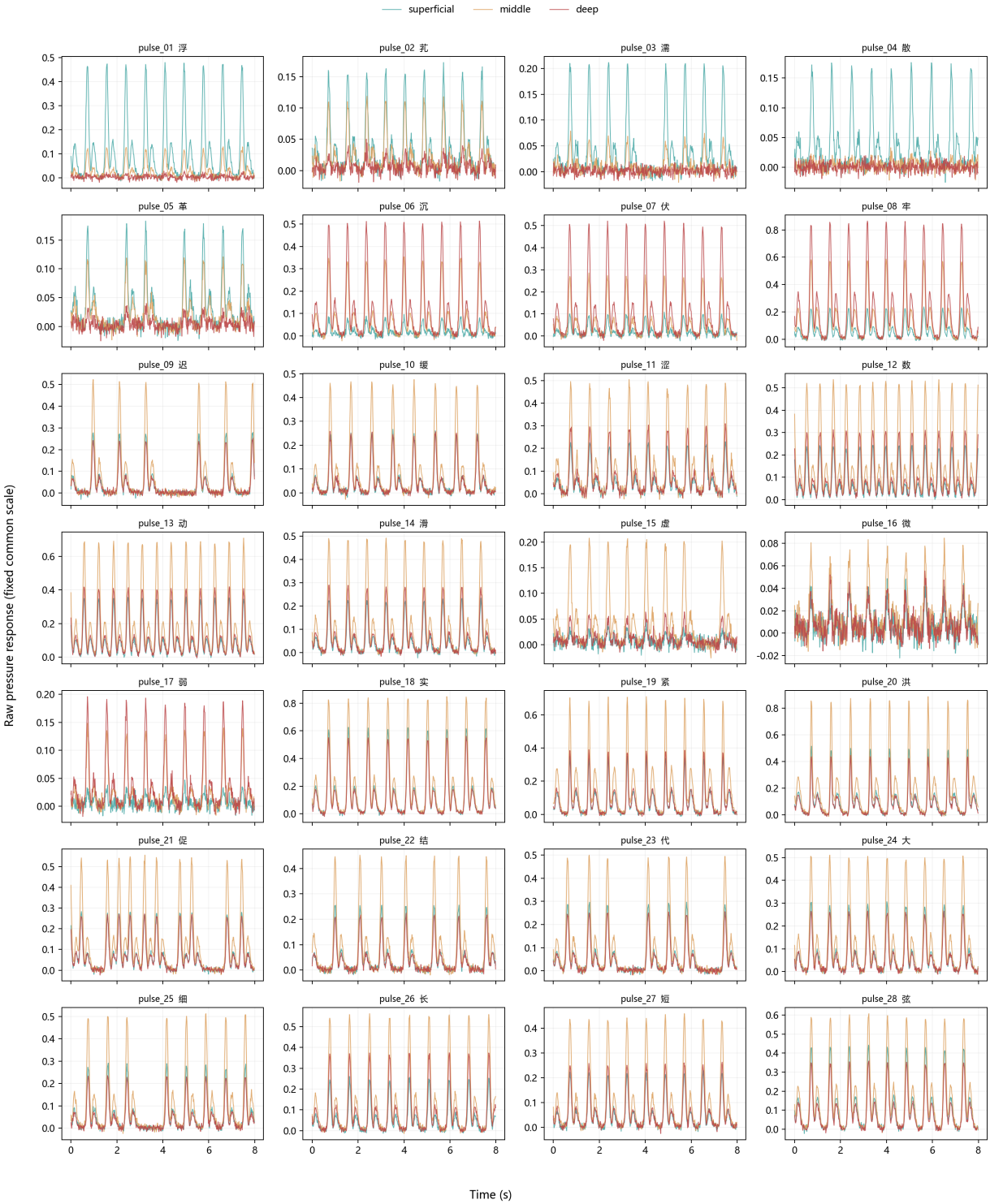


Figure 1: Engineering visualization of the 28 operational pulse prototypes at superficial, middle, and deep pressure. The shared vertical scale preserves force differences. The figure audits the synthetic renderer; it is not clinical validation.